SQL_TEST_01 · BUILD 3 · HANDOFF BRIEF

Semantic Search & Retrieval

A standalone, in-progress reference for Build 3 — Phases 1-2 in full, every file, the full verification log, and a dedicated look at how a question actually flows through the agent at runtime.

STATUS	COMMITS	CONTINUES	DATA CHOICE
Phases 1-2 complete, in progress	`2e36f06` → `f4f2eb4`	Build 2 / Build 2.5	clean.allocations.summary

What Was Built, and Why

Build 2 gave the agent structured data to read. Build 3 gives it **semantic** reach over the one genuinely unstructured field in the schema — `clean.allocations.summary`, free-text equipment descriptions — so questions about the general kind or theme of a checkout ("camera gear") work even without exact keyword matches.

Data choice was deliberated, not assumed: two alternatives (a new purpose-sourced document corpus, and a meta/self-referential RAG agent over this project's own documentation) were considered and explicitly deferred — the documentation angle is logged as a future **Build 3.5**. `summary` won because it extends the same dataset across builds rather than sprawling into a new kind of corpus, per this project's own stated principle.

No new database role was needed here, unlike Build 2.5 — semantic search is just another *read*, so it uses `appdb_reader` like every other query tool.

1 Enable Semantic Search

Goal: get `pgvector` running, embed the existing corpus, and prove raw similarity search works before touching the agent at all.

> **Infrastructure change:** `docker-compose.yml`'s `postgres` image swapped from `postgres:17` to `pgvector/pgvector:pg17` — same Postgres 17, extension pre-installed. Data survives the swap (it lives in the separate `pgdata` volume, not the image). The swap did trigger a collation-version warning (the new image bundles a different glibc/ICU version) — resolved with `ALTER DATABASE appdb REFRESH COLLATION VERSION;`, safe for this dataset's plain ASCII/business text.

`008_add_summary_embedding.sql`

```sql
-- adds a pgvector embedding column for clean.allocations.summary.
-- Depends on the vector extension (Phase 1, step 1) and clean.allocations existing.

ALTER TABLE clean.allocations
    ADD COLUMN IF NOT EXISTS summary_embedding vector(384);
```

> **Embedding model choice:** `sentence-transformers` (`all-MiniLM-L6-v2`, 384-dim), run locally — no new paid API/account, zero marginal cost, sufficient quality for equipment-description text. Considered and declined: Voyage AI (Anthropic's own recommended embeddings partner) and OpenAI's `text-embedding-3-small` — both good options, both would have added a new credential for a task that doesn't need frontier-tier embeddings.

`embed_summaries.py`

```python
import os
from dotenv import load_dotenv
import psycopg
from pgvector.psycopg import register_vector
from sentence_transformers import SentenceTransformer

load_dotenv(encoding="utf-8-sig")

model = SentenceTransformer("all-MiniLM-L6-v2")

conn = psycopg.connect(
    host=os.environ.get("POSTGRES_HOST", "127.0.0.1"),
    port=5432,
    dbname=os.environ["POSTGRES_DB"],
    user=os.environ["POSTGRES_USER"],
    password=os.environ["POSTGRES_PASSWORD"],
)
```

```
register_vector(conn)

with conn.cursor() as cur:
    cur.execute("SELECT allocation_id, summary FROM clean.allocations")
    rows = cur.fetchall()

texts = [summary if summary else "" for _, summary in rows]
embeddings = model.encode(texts, show_progress_bar=True)

with conn.cursor() as cur:
    cur.executemany(
        "UPDATE clean.allocations SET summary_embedding = %s WHERE allocation_id = %s",
        [(embedding, allocation_id) for (allocation_id, _), embedding in zip(rows, embeddings)],
    )
conn.commit()
conn.close()
```

Verified: all 1,535 rows embedded, correct 384 dimensions, self-distance = 0 for spot-checked rows (a vector's distance to itself must always be zero — confirms the vector math itself, not just non-null values).

```
search_summaries.py — standalone verification script

query_embedding = model.encode(query_text)
cur.execute("""
    SELECT allocation_id, summary, summary_embedding <=> %s AS distance
    FROM clean.allocations
    ORDER BY distance
    LIMIT 5
""", (query_embedding,))
```

> **Relevance checkpoint passed via contrast test:** `"camera equipment"` scored `0.469`-`0.513`, returning genuinely coherent video/audio production gear. A deliberately out-of-domain query, `"office desk chair"` (this corpus has zero office furniture), scored meaningfully worse at `0.657`-`0.773` — real discrimination, even though the office-chair query's own top-5 results were still nonsense (there's nothing relevant to find; a plain top-K query has no "no good match" concept). That gap became the seed for Phase 2's threshold work.

2 Wire Into the Agent

Goal: expose semantic search as a real tool, with an actual relevance threshold instead of always returning top-K regardless of quality.

> **Calibration methodology correction, mid-phase:** a first calibration attempt sampled real corpus summaries as "in-domain queries" and got a misleadingly large separation (`0.16` vs `0.65`) — because many real summaries are near-duplicates of each other (the same standard kit checked out repeatedly), this measured duplication, not relevance. Using that number as the threshold would have rejected Phase 1's own working example (`"camera equipment"` at `0.469`-`0.513` is *higher* than the flawed `0.4066` midpoint). Recalibrated using short natural-language queries on **both** sides — matching how the agent actually queries — giving a real, narrower, correctly-placed separation.

```
calibrate_threshold.py — corrected methodology
in_domain_queries = [
    "camera equipment", "video tripod", "audio recording gear", "lighting kit",
    "memory card", "laptop charger", "camera lens", "microphone", "HDMI cable",
    # ... 20 total, short phrases for things this corpus genuinely has
]
out_of_domain_queries = [
    "office desk chair", "kitchen blender", "garden shovel", "running shoes",
    # ... 20 total, short phrases for things this corpus has none of
]
# For each query: embed it, find its single nearest neighbor's distance in the corpus.
# Result: in-domain 0.3109-0.6019 (mean 0.4519) vs out-of-domain 0.6538-0.8235 (mean 0.7584)
# Clean separation - midpoint threshold ~0.6279, rounded to 0.63
```

```
tools.py — search_summaries
_embedding_model = None

def _get_embedding_model():
    global _embedding_model
    if _embedding_model is None:
        _embedding_model = SentenceTransformer("all-MiniLM-L6-v2")
    return _embedding_model

SIMILARITY_DISTANCE_THRESHOLD = 0.63

def search_summaries(query_text: str, limit: int = 5) -> list[dict]:
    model = _get_embedding_model()
    query_embedding = model.encode(query_text)
    # ... connects as appdb_reader, register_vector(conn) ...
    cur.execute("""
        SELECT allocation_id, summary, summary_embedding <=> %s AS distance
```

```
        FROM clean.allocations
        WHERE summary_embedding <=> %s < %s
        ORDER BY distance
        LIMIT %s
    """, (query_embedding, query_embedding, SIMILARITY_DISTANCE_THRESHOLD, limit))
    rows = cur.fetchall()

    if not rows:
        return [{"message": f"No sufficiently relevant results for {query_text!r} ..."}]
    return [{"allocation_id": aid, "summary": summary, "distance": round(float(dist), 4)} for aid,
summary, dist in rows]
```

Real incident during agent wiring: the `elif block.name == "search_summaries":` dispatch branch was missed when adding the tool schema. Because Python's `if/elif` chain had no matching branch and no `else`, the loop variable `result` silently kept its value from the *previous* tool call (`describe_table`'s column list) — no exception, so nothing looked broken. The model received a mismatched result and pragmatically fell back to a plain SQL `ILIKE '%camera%'` query instead, producing a correct-looking final answer that never actually exercised `search_summaries` at all. Caught only by checking that the tool's actual output matched its own logic, not by the final answer looking plausible.

agent.py — the fix

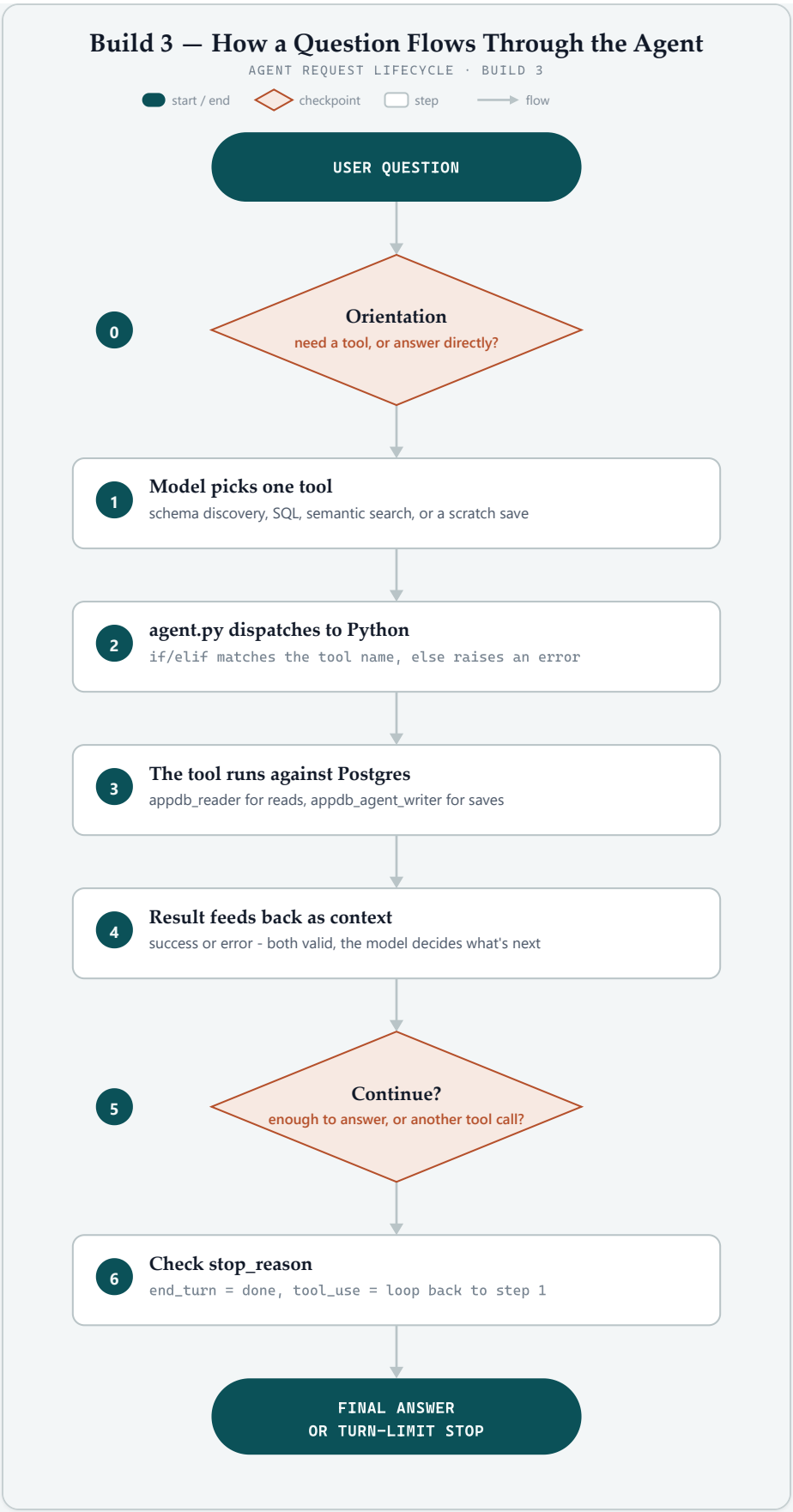
```
elif block.name == "search_summaries":
    result = search_summaries(block.input["query_text"])
else:
    raise ValueError(f"Unknown tool name: {block.name}")
```

That `else` is the generalizable part of the fix — any future tool added without its dispatch branch now fails loudly, instead of silently reusing stale data from whatever ran before it.

Verified live, twice: first pass (before the fix) proved the bug — the printed "result" for `search_summaries` was actually `describe_table`'s stale output. Second pass (after the fix) showed the real thing: a model-loading progress bar (first genuine invocation lazy-loads the embedding model) followed by the correct `{allocation_id, summary, distance}` shape, matching Phase 1's own numbers exactly for the same query. The agent then combined those results with a follow-up `run_sql_query` for exact checkout dates — a two-tool combination nobody coded explicitly, just the model composing what the system prompt made available.

How a Question Flows Through the Agent

A dedicated walkthrough of the runtime request lifecycle as it stands after Phase 2 — distinct from the build-phase diagrams above, this is the architecture of a single question's journey, every time.



Same visual system as the build/phase diagrams — this one documents runtime behavior, not a build plan.

1. The question enters as a single message. `user_question` is wrapped into `messages = [{"role": "user", "content": user_question}]` — that's the entire starting context. The model gets no hardcoded schema, no pre-fetched data — just the question, the `SYSTEM_PROMPT`, and the five tool schemas (`run_sql_query`, `list_tables`, `describe_table`, `save_dataframe`, `search_summaries`).

2. First API call — the model orients itself. Per the system prompt ("you don't know the schema in advance"), it typically calls `list_tables` first, then `describe_table` on whatever looks relevant — this is pure discovery, no data yet. Each of these returns as a `tool_use` block; the `while turn < MAX_TOOL_TURNS` loop dispatches it through the `if/elif/else` chain in `agent.py`, gets a result back from `tools.py`, and feeds that result back to the model as the next turn's context.

3. The real judgment call — which data tool to reach for. This is decided by the model itself, guided by the system-prompt line: "*For questions about the general kind or theme of equipment... use search_summaries instead of writing SQL yourself.*" Nothing in the code forces this choice — it's the model reading that instruction and applying it. A structured question ("how many allocations did GART make?") gets `run_sql_query`. A thematic one ("what camera equipment...") gets `search_summaries`.

4. If it picks `search_summaries`: `tools.py` embeds the query text with the lazily-loaded `sentence-transformers` model (loaded once, cached module-level), runs a cosine-distance query against `clean.allocations.summary_embedding` filtered by the calibrated `0.63` threshold, and returns either real `{allocation_id, summary, distance}` rows or an explicit "no sufficiently relevant results" message — never a silent empty list.

5. Observed emergent pattern — semantic search feeds structured follow-up. In both live tests, the model didn't stop at `search_summaries`'s results — it took the returned `allocation_id`s and issued a *second*, precise `run_sql_query` (e.g., pulling exact `actual_start`/`actual_end` dates) to enrich the answer. Nobody coded that combination explicitly; it's the model composing two tools because the system prompt gave it both options and the question benefited from both.

6. Self-correction, observed again. When a query is written as `FROM allocations` (unqualified) instead of `FROM clean.allocations`, it fails with a real Postgres error — which comes back as a normal (non-crashing) `tool_result` with `is_error: True`, per Build 1's hardening pass. The model reads that error and retries schema-qualified, without any help.

7. Termination. The loop ends when `stop_reason` is `end_turn` (a plain-text final answer, no more tool calls) — or, as a safeguard, when `MAX_TOOL_TURNS` (10) is hit, printing a clean stop message instead of hanging forever.

Underneath all of this, the same trust boundary from Build 1 still governs everything, now with two lanes: the LLM proposes actions but never executes anything directly → `agent.py`'s dispatch is the gate (now with the `else: raise` guard against an unhandled tool silently reusing stale data) → the database role is the real backstop. Read paths (`run_sql_query`, `list_tables`, `describe_table`, `search_summaries`) all run as `appdb_reader` — physically incapable of writing. `save_dataframe` alone runs as `appdb_agent_writer` — physically incapable of touching anything outside `agent_scratch`. The

model's own judgment decides *which* tool to call; the roles decide what's *possible* regardless of what it decides.

Verification Log

- ✓ **Embedding backfill**
  1,535/1,535 rows embedded, correct 384 dimensions, self-distance = 0 on spot checks.
- ✓ **Relevance contrast test**
  `"camera equipment"` (0.469-0.513) vs `"office desk chair"` (0.657-0.773) — real, meaningful separation.
- ✓ **Threshold calibration**
  Corrected methodology: in-domain 0.3109-0.6019 vs out-of-domain 0.6538-0.8235 → threshold 0.63, empirically justified not guessed.
- ✓ **search_summaries direct test**
  Relevant query returns real rows; out-of-domain query returns the explicit "no relevant results" message, not an empty list.
- ✓ **Agent integration, twice**
  First run exposed the missing-dispatch bug (stale result silently reused); second run, post-fix, showed the real tool executing (progress bar + matching distances) and combining with a follow-up SQL query for exact dates.

Known Limitations & Open Items

Resolved — automated re-embedding for new rows. `embed_summaries.py`'s query changed from an unconditional `SELECT allocation_id, summary FROM clean.allocations` to `... WHERE summary_embedding IS NULL`, with an early exit printing `"No rows need embedding - everything is already up to date."` when there's nothing to do. Verified live: ran once with all 1,535 rows already embedded (no-op, correct message); then manually nulled two real rows' embeddings to simulate new rows arriving via a future pipeline re-run, re-ran the script, and confirmed only those 2 rows were re-embedded (not all 1,535), with both landing back at 384 dimensions and self-distance 0. No dedicated pytest test was added for this specific behavior — it mirrors how `ingest.py/transform.py` are verified (live round-trip, not unit-tested internals), and the existing `test_all_rows_have_embeddings_with_correct_dimensions` test already guards against a future regression leaving rows unembedded.

Resolved — real vector index for scale. `search_summaries`' cosine-distance query originally had no index to use, meaning every call did a full sequential scan of `clean.allocations` — fine at 1,535 rows, but not something that scales. `009_add_summary_embedding_hnsw_index.sql` adds `CREATE INDEX ... USING hnsw (summary_embedding vector_cosine_ops)`, matching the `<=>` operator already in use. Verified live via `EXPLAIN ANALYZE`: the query planner picks the new index for an `Index Scan` even at the current small scale, rather than falling back to a sequential scan. Codified as `test_hnsw_index_exists_on_summary_embedding` (queries `pg_index/pg_class/pg_am` directly) — full suite now 14/14 passing.

The 0.63 threshold is an empirically-grounded heuristic, not a guarantee. Calibrated from 40 curated queries (20 in-domain, 20 out-of-domain) against this specific corpus - a reasonable, evidence-based starting point, but not something that generalizes automatically if the corpus or query style changes substantially.

Carried forward from Build 2.5: the `list_tables/describe_table` merge-into-one-call idea and the lack of a per-table row/size cap on `agent_scratch` both remain low-priority, unaddressed items.

Incidental fix during this build: `profile.py` was renamed to `profile_raw_data.py` after it was found silently shadowing Python's stdlib `profile` module, breaking an unrelated import deep in `torch`'s own import chain. Unrelated to semantic search itself, logged separately in project memory.

Environment State

COMPONENT	DETAIL
Postgres image	`pgvector/pgvector:pg17 (was postgres:17)`
New extension	`vector`
New column	`clean.allocations.summary_embedding vector(384)`
New SQL files	`008_add_summary_embedding.sql, 009_add_summary_embedding_hnsw_index.sql`
Embedding model	`sentence-transformers, all-MiniLM-L6-v2 (local, free)`
New/changed Python	`embed_summaries.py (+idempotent re-embedding), search_summaries.py, calibrate_threshold.py, tools.py (+search_summaries), agent.py (+tool, +dispatch, +else guard, +prompt line), test_semantic_search.py (4 tests)`
Renamed	`profile.py → profile_raw_data.py`
Role used	`appdb_reader (no new role needed for this build)`

In progress — Phases 1-2 complete and verified. This brief will be updated if further Build 3 phases happen, the same way the Build 2.5 brief was updated after its table-cap fix. Ongoing cross-build command reference lives in `sql-test-01-cheatsheet.html`.